Chapter 5—Linkage, Recombination, and the Mapping of Genes on Chromosomes

Fill in the Blank

1. Genes that "travel" together from one generation to the next more often than not exhibit ______________________ _______________________.
   Ans: genetic linkage
   Difficulty: 1

2. A probability test that measures "goodness of fit" between observed and predicted results is a __________ ________________ test.
   Ans: chi square test
   Difficulty: 1

3. A hypothesis that predicts no linkage between genes is called the ______________ hypothesis.
   Ans: null hypothesis
   Difficulty: 2

4. Genes that can serve points of reference on a chromosome are useful as genetic markers while cytologically visible abnormalities that also make it possible to keep track of chromosomes are called ________________________ markers.
   Ans: physical
   Difficulty: 1

5. The movement of chiasmata toward the end of a chromosome is called ______________________.
   Ans: teminalization
   Difficulty: 2

6. One percentage point of recombination, or recombination frequency, is a unit of measure called either centimorgan or ___________ _____________.
   Ans: map unit
   Difficulty: 1

7. Crossovers do not occur independently and the occurrence of one crossover reduces the likelihood of another occurring elsewhere on the chromosome this is called __________________________ __________________________.
   Ans: chromosomal interference
   Difficulty: 2

8. A tetrad that contains four recombinant class haploid cells is known as a ____________________ ditype.
   Ans: non-parental ditype
   Difficulty: 1

9. A tetrad that carries four kinds of haploid cells: two different parental and two different recombinant class spores is referred to as ________________________.
   Ans: tetratype
   Difficulty: 1

10. Mistakes in mitosis during development often result in ______________________ organisms containing tissues of different genotypes.
    Ans: mosaic
    Difficulty: 1

Multiple Choice

11. The R/r and S/s genes are linked and 10 map units apart. In the cross Rs/rS × rs/rs what fraction of the progeny will be RS/rs?
    A) 5%
    B) 10%
    C) 25%
    D) 40%
    E) 45%
    Ans: A
    Difficulty: 3

12. If the map distance between genes A and B is 10 map units and the map distance between genes B and C is 25 map units, what is the map distance between genes A and C?
    A) 15 map units
    B) 35 map units
    C) Either 15 map units or 35 map units, depending on the order of the genes.
    D) The map distance between A and C can not be predicted from these data.
    Ans: C
    Difficulty: 2

13. In *Drosophila*, singed bristles (sn) and cut wings (ct) are both caused by recessive, X-linked alleles. The wild type alleles (sn^+ and ct^+) are responsible for straight bristles and intact wings, respectively. A female homozygous for sn and ct^+ is crossed to a sn^+ct male. The F1 flies are interbred. The F2 males are distributed as follows:

sn ct	13
sn ct^+	36
sn^+ ct	39
sn^+ ct^+	12

What is the map distance between sn and ct?

A) 12 m.u.
B) 13 m.u.
C) 25 m.u.
D) 50 m.u.
E) 75 m.u.

Ans: C
Difficulty: 2

The following information applies to problems 14 & 15. In *Drosophila*, singed bristles (sn) and carnation eyes (car) are both caused by recessive, X-linked alleles. The wild-type alleles (sn^+ and car^+) are responsible for straight bristles and red eyes, respectively. A sn car female is mated to a sn^+ car^+ male and the F1 progeny are interbred. The F2 are distributed as follows:

sn car	55
sn car^+	45
sn^+ car	45
sn^+ car^+	55
	200

14. What is the value of X^2 for a test of the hypothesis that the sn and car genes are unlinked?

A) 0.5
B) 1.0
C) 2.0
D) 0.4
E) 20

Ans: C
Difficulty: 2

15. What is the p value from this test? (Pick the most accurate choice.)
A) $p>.5$
B) $.5>p>.1$
C) $p<.1$
D) $p<.05$
E) $p<.01$
Ans: A
Difficulty: 3

16. Suppose the L and M genes are on the same chromosome but separated by 100 map units. What fraction of the progeny from the cross LM/lm × lm/lm would be Lm/lm?
A) 10%
B) 25%
C) 50%
D) 75%
E) 100%
Ans: B
Difficulty: 3

17. The pairwise map distances for four linked genes are as follows: A-B = 22 m.u., B-C = 7 m.u., C-D = 9 m.u., B-D = 2 m.u., A-D = 20 m.u., A-C = 29 m.u. What is the order of these four genes?
A) ABCD
B) ADBC
C) ABDC
D) BADC
E) CADB
Ans: B
Difficulty: 3

18. The zipper-like connection between paired homologs in early prophase is known as a:
A) spindle fiber.
B) synaptic junction.
C) synaptonemal complex.
D) chiasma.
E) none of the above
Ans: C
Difficulty: 1

19. The measured distance between genes D and E in a two point test cross is 50 map units. What does this mean in physical terms?
A) D and E are on different pairs of chromosomes.
B) D and E are linked and exactly 50 map units apart.
C) D and E are linked and at least 50 map units apart.
D) either a or b
E) either a or c
Ans: E
Difficulty: 2

20. Dihybrid test crosses are made between each pairwise combination of the three genes O, P and Q with the following results:

OoPp × oopp		PpQq × ppqq		OoQq × ooqq	
OP	152	PQ	88	OQ	198
Op	348	Pq	415	Oq	289
oP	353	pQ	413	oQ	311
op	147	pq	84	oq	202
	1000		1000		1000

You wish to use the X^2 test to determine whether the O and P genes are linked. Which set of data would you use?
A) the first cross
B) the second cross
C) the third cross
D) all three crosses
E) the first and second crosses
Ans: A
Difficulty: 1

The following information applies to problems 21 & 22. In humans, the genes for red-green color blindness (R=normal, r=color-blind) and hemophilia A (H=normal, h=hemophilia) are both X-linked and only 3 map units apart.

21. Suppose a woman has four sons, and two are colorblind but have normal blood clotting and two have hemophilia but normal color vision. What is the probable genotype of the woman?
A) HR/hr
B) Hr/hr
C) hr/hR
D) Hr/hR
E) HR/Hr
Ans: D
Difficulty: 3

22. A woman whose mother is colorblind and whose father has hemophilia A is pregnant with a boy and wants to know the probability that he will have normal vision and blood clotting. What is the probability?
A) .03
B) .15
C) .485
D) .47
E) .015
Ans: E
Difficulty: 4

23. The R/r and S/s genes are linked and 10 map units apart. In the cross Rs/rS × rs/rs what percentage of the progeny will be Rs/rs?
A) 5%
B) 10%
C) 25%
D) 40%
E) 45%
Ans: E
Difficulty: 3

The following information applies to problems 24-27. In fruit flies, the recessive pr and cn mutations cause brown and bright-red eyes, respectively (wild-type flies have brick-red eyes). The double mutant pr cn combination has orange eyes. A female who has wild-type eyes is crossed to an orange-eyed male. Their progeny have the following distribution of eye colors:

wild-type	8
brown	241
bright-red	239
orange	12
	500

24. Which classes are the parental types?
A) wild-type and orange
B) brown and bright-red
C) wild-type and brown
D) bright-red and orange
E) there is no way to determine this
Ans: A
Difficulty: 2

25. What is the genotype of the mother of these progeny?
A) $prcn/pr^+cn^+$
B) pr^+cn/pr^+cn
C) $pr^+cn/prcn^+$
D) $prcn^+/prcn^+$
E) prcn/prcn
Ans: C
Difficulty: 2

26. The mother of these progeny resulted from a cross between two flies from true breeding lines. What are the genotypes of these two lines?
A) $prprcn^+cn^+$ and pr^+pr^+cncn
B) $pr^+pr^+cn^+cn^+$ and prprcncn
C) pr^+prcn^+cn and prprcncn
D) $pr^+prcncn$ and $prprcn^+cn$
E) either a or b could be true
Ans: A
Difficulty: 2

27. What is the map distance between the pr and cn genes?
A) 20 m.u.
B) 2 m.u.
C) 4 m.u.
D) 46 m.u.
E) 8 m.u.
Ans: C
Difficulty: 2

28. A dihybrid test cross is made between genes H and I. Four categories of offspring are produced: HI, Hi, hI, and hi. You wish to use the X^2 test to test the hypothesis that the H and I genes are unlinked. How many degrees of freedom would there be in this test?
A) 1
B) 2
C) 3
D) 4
E) 0
Ans: C
Difficulty: 1

29. Which of the following processes can generate recombinant gametes?
A) Segregation of alleles in a heterozygote.
B) Crossing over between two linked heterozygous loci.
C) Independent assortment of two unlinked heterozygous loci.
D) b and c
E) a, b and c
Ans: D
Difficulty: 2

30. Crossing over takes place in paired bivalents consisting of ______ chromatids, and involves _______ of the chromatids.
A) 2, 2
B) 2, 4
C) 4, 2
D) 4, 4
E) 8, 4
Ans: C
Difficulty: 2

31. In *Drosophila*, the genes y (yellow body) and car (carnation eyes) are located at opposite ends of the X chromosome. In doubly heterozygous females (y^+ car^+/y car), a single chiasma is observed somewhere along the X chromosome in 90% of the examined oocytes. No X chromosomes with multiple chiasmata are observed. What percentage of the male progeny from such a female would be recombinant for y and car?
A) 5%
B) 10%
C) 45%
D) 55%
E) 90%
Ans: C
Difficulty: 4

32. Genes Q and R are 20 map units apart. If a plant of genotype QR/qr is selfed, what percentage of the progeny will be rs in phenotype?
A) 4%
B) 10%
C) 16%
D) 20%
E) 40%
Ans: C
Difficulty: 4

33. The map of a chromosome interval is Q 10 R 40 S. Which interval would likely show the higher ratio of double to single chiasmata?
A) Q-R
B) R-S
C) The ratios would be the same in the two intervals.
D) Two chiasmata never occur in the same interval.
Ans: B
Difficulty: 3

34. The map of a chromosome interval is A 10 B 40 C. From the cross Abc/aBC × abc/abc, how many double crossovers would be expected out of 1000 progeny?
A) 5
B) 10
C) 20
D) 40
E) 80
Ans: D
Difficulty: 3

35. The cross Lpq/lPQ × lpq/lpq is carried out, the L gene is found to be in the middle. What would be the genotypes of the double crossover gametes in this cross?
A) LPQ and lpq
B) LpQ and lPq
C) lpQ and LPq
D) Lpq and lPQ
E) cannot be determined
Ans: A
Difficulty: 3

36. Suppose a three-point testcross was conducted involving the genes X, Y and Z. If the most abundant classes are XYz and xyZ and the rarest classes are xYZ and Xyz, which gene is in the middle?
A) X
B) Y
C) Z
D) cannot be determined
Ans: B
Difficulty: 2

37. In *Drosophila*, the genes b, c and sp are linked and in the order given, the distances being b-c = 30 m.u. and c-sp = 20 m.u. These intervals exhibit 90% interference. How many double crossovers would be recovered in a three-point cross involving b, c and sp out of 1000 progeny?
A) 3
B) 6
C) 54
D) 60
E) 600
Ans: B
Difficulty: 4

38. In tetrad analysis, the criterion for linkage of two genes is:
A) NPD = T.
B) PD = T.
C) PD = NPD.
D) PD > NPD.
E) PD > T.
Ans: D
Difficulty: 3

39. In tetrad analysis, NPD asci result from:
A) independent assortment of unlinked genes.
B) double crossovers between linked genes.
C) single crossovers between linked genes.
D) single crossovers between a gene and a centromere.
E) a or b
Ans: E
Difficulty: 3

40. In tetrad analysis, second-division segregations result from:
A) single crossovers between linked genes.
B) double crossovers between linked genes.
C) single crossovers between a gene and a centromere.
D) independent assortment of unlinked genes.
E) nondisjunction of homologs.
Ans: C
Difficulty: 2

41. Tetrad analysis shows that crossing over occurs at the four-strand stage (i.e., after replication) because, when two genes are linked,:
A) NPD > T
B) T > NPD
C) T > PD
D) PD > NPD
E) PD > T
Ans: B
Difficulty: 4

The following information applies to problems 42 & 43. Consider a pair of homologous chromosomes heterozygous for three genes (e.g. ABC/abc) during prophase I of meiosis. Let the sister chromatids of one homolog be numbered 1 and 2; and the sister chromatids of the other homolog be numbered 3 and 4.

42. A crossover that would result in genetic recombination (e.g., Abc or aBC) could involve which chromatids?
A) 1 & 2 or 3 & 4
B) 1 & 3 or 2 & 4
C) 1 & 4 or 2 & 3
D) 1 & 3 or 1 & 4 or 2 & 3 or 2 & 4
E) any two of the four chromatids
Ans: D
Difficulty: 3

43. Assume a double crossover occurs in this pair of chromosomes that results in chromatids of the genotype AbC and aBc. If the first crossover (the one between A and B) involves chromatids 1 & 4, which chromatids could be involved in the second crossover?
A) 1 & 2 or 3 & 4
B) 1 & 3 or 2 & 4
C) 1 & 4 or 2 & 3
D) 1 & 3 or 1 & 4 or 2 & 3 or 2 & 4
E) any two of the four chromatids
Ans: D
Difficulty: 4

44. Sturtevant's detailed mapping studies of the X chromosome of *Drosophila* established what genetic principle?
A) That genes are arranged in a linear order on the chromosomes.
B) That genes are carried on chromosomes.
C) That sex determination is controlled by the X and Y chromosomes.
D) That segregation of an allelic gene pair is accompanied by disjunction of homologous chromosomes.
E) That different pairs of chromosomes assort independently.
Ans: A
Difficulty: 2

45. The map of a chromosome is A 40 B 10 C 10 D 40 E. Suppose an investigator were to conduct two trihybrid crosses, cross 1: ACE/ace × ace/ace and cross 2: BCD/bcd × bcd/bcd with the purpose of comparing the levels of crossover interference between the two crosses. Which cross would be expected to have the higher coefficient of coincidence?
A) cross 1
B) cross 2
C) Neither, the coefficient of coincidence is constant along a chromosome.
D) There is no way to predict which cross would have the higher coefficient of coincidence.
Ans: A
Difficulty: 4

46. Suppose an individual is heterozygous for a pair of alleles (e.g. A/a). Under what conditions would a crossover in a somatic cell of this individual lead to a clone of cells homozygous for a? (Pick the most precise answer.)
A) The crossover would have to occur between the A locus and the centromere and involve two homologous (non-sister) chromatids.
B) The crossover would have to occur between the A locus and the end of the chromosome and involve two homologous (non-sister) chromatids.
C) The crossover would have to occur on the same chromosome arm as the A locus and involve two homologous (non-sister) chromatids.
D) The crossover would have to occur on the same chromosome as the A locus and involve two homologous (non-sister) chromatids.
E) The crossover would have to occur between the A locus and the centromere and involve two sister chromatids (not homologous) chromatids.
Ans: A
Difficulty: 3

47. Suppose an individual is heterozygous for two linked pairs of alleles on the same chromosome arm, Ab/aB such that the A locus is closer to the centromere than the B locus. Under what conditions would a crossover in a somatic cell generate a twin spot, i.e. two adjacent clones of cells, one clone homozygous for a and the other clone homozygous for b?

A) The crossover would have to occur between the A locus and the centromere locus and involve two homologous (non-sister) chromatids.
B) The crossover would have to occur between the A locus and the B locus locus and involve two homologous (non-sister) chromatids.
C) The crossover would have to occur between the B locus and the end of the chromosome locus and involve two homologous (non-sister) chromatids.
D) A double crossover would have to occur, with one crossover between the A locus and the centromere and a second crossover between the A and B loci locus and both crossovers would have to involve two homologous (non-sister) chromatids.
E) No crossover in a somatic cell could generate a twin spot.

Ans: A
Difficulty: 4

48. Individuals heterozygous for the RB^+ and RB^- alleles can develop tumors as a result of:

A) a mitotic crossover that leads to homozygosity for both RB^+ and RB^-.
B) a somatic mutation in the RB^+ allele that leads to homozygosity for RB^-.
C) a somatic mutation in the RB^- allele that leads to homozygosity for RB^+.
D) the fact that RB^- is dominant to RB^+.
E) a and b

Ans: E
Difficulty: 3

49. What happens physically during the process of crossing over?

A) Two homologous chromatids break and rejoin at random sites along the chromosome.
B) The genetic information on one chromatid is replaced by copying genetic information from a homologous chromatid without there being any physical exchange between the chromosomes.
C) Two homologous chromatids break and rejoin at precisely the same site along the chromosome so that there is no loss or gain of material on either product.
D) It is not known what occurs during crossing over.

Ans: C
Difficulty: 2

50. Here is a list of events during meiosis I. It is not in the correct order.
    A) Homologous chromosomes are roughly aligned but not physically linked.
    B) Homologous chromosomes segregate to opposite poles.
    C) Homologous chromosomes are linked by synaptonemal complexes.
    D) Homologous chromosomes are linked by chiasmata.
    E) Chromosomes replicate.
    What is the correct order of these events?
A) ACDBE
B) AECDB
C) EACDB
D) EADCB
E) CDABE
Ans: C
Difficulty: 3

51. Some of the larger human chromosomes typically contain multiple chiasmata during meiotic prophase. If you were to carefully study the distribution of these chiasmata, what would you find?
A) Chiasmata are randomly distributed along chromosomes.
B) All chromosome pairs have the same number of chiasmata.
C) A single chromosome pair always has the same number of chiasmata in every meiotic cell.
D) Chiasmata are spaced along a chromosome arm more regularly than would be expected by chance.
E) Chiasmata are spaced more irregularly along a chromosome arm than would be expected by chance.
Ans: D
Difficulty: 4

Matching

In a mating between haploid yeast cells of type *a* = *his4/TRP1* × type " = *HIS4/trp1*, *a*/" diploid offspring result. Match the appropriate ditype that results when these undergo meiosis with the genotypes shown: parental (PD), non-parental (NPD), or tetratype (T).

52. ______ *his4/TRP1; his4/trp1; HIS4/trp1; HIS4/TRP1*
Ans: PD
Difficulty: 1

53. ______ *his4/TRP1; his4/TRP1; HIS4/trp1; HIS4/trp1*
Ans: T
Difficulty: 1

54. ______ *his4/trp1; his4/trp1; HIS4/TRP1;HIS4/TRP*
Ans: NPD
Difficulty: 1

55. ______ *HIS4/trp1; HIS4/TRP1*; *his4/TRP1; his4/trp1*
Ans: T
Difficulty: 1

56. ______ *HIS4/trp1; HIS4/trp1; his4/TRP1; his4/TRP1*
Ans: PD
Difficulty: 1

Match the following results of offspring in an ordered octad with the appropriate segregation pattern: first-division (1) or second division (2).

57. ______ *ws; ws; ws; ws; ws+; ws+; ws+; ws+*
Ans: 1
Difficulty: 1

58. ______ *ws; ws; ws+; ws+; ws; ws; ws+; ws+*
Ans: 2
Difficulty: 1

59. ______ *ws; ws; ws; ws; ws+; ws+; ws; ws*
Ans: 1
Difficulty: 1

60. ______ *ws+; ws+; ws+; ws+; ws; ws; ws; ws*
Ans: 2
Difficulty: 1

61. ______ *ws+; ws+; ws; ws; ws+; ws+; ws; ws*
Ans: 2
Difficulty: 1

True or False

62. Recombination due to crossing over occurs only rarely in mitosis.
Ans: True
Difficulty: 1

63. Two genes are considered linked when F_2 progeny more commonly show the recombinant genotype.
Ans: False
Difficulty: 1

64. When considering the chi-square test, a p-value of 0.05 is often considered significant but is actually an arbitrary assignment of significance.
Ans: True
Difficulty: 2

65. Chiasmata are structures of cross over between sister chromatids of homologous chromosomes.
Ans: False
Difficulty: 1

66. Chiasmata can be seen through a light microscope and are sites of recombination.
Ans: True
Difficulty: 1

67. Three-point crosses are more tedious and less accurate than two-point crosses.
Ans: False
Difficulty: 1

68. Chromosomal interference is not uniform and may significantly vary within a region of a single chromosome.
Ans: True
Difficulty: 2

69. Genetic mapping distance calculated for a chromosome is directly related to physical distance along that chromosome.
Ans: False
Difficulty: 2

70. Genes chained together by linkage relationships are known collectively as a linkage group.
Ans: True
Difficulty: 2

71. Ascospores are haploid cells that can germinate, replicate by mitosis, and survive as haploid individuals.
Ans: True
Difficulty: 1

Short Answer

72. C. Stern studied recombination between two homologous X chromosomes in *Drosophila* in which one chromosome had two cytologically visible abnormalities at opposite ends. What did he find?
    Ans: Crossover products determined genetically were also recombinant for the cytological markers, indicating that genetic recombination involves physical exchange between homologs.
    Difficulty: 3

73. What is a chiasma?
    Ans: The cytological manifestation of a crossover.
    Difficulty: 1

74. When setting up crosses to determine map distances, why do geneticists prefer to cross the hybrid individuals to individuals homozygous for the recessive alleles in the cross?
    Ans: Because the recessive alleles from the tester parent will allow the alleles from the other parent to be revealed, so that all genotypes will be apparent from the phenotypes in the progeny.
    Difficulty: 1

75. What is a testcross?
    Ans: A cross in which one parent is homozygous for the recessive alleles.
    Difficulty: 1

76. What feature of meiosis in fungi such as Neurospora makes them especially suitable for the analysis of recombination?
    Ans: All of the products of a single meiosis remain packaged together in an ascus.
    Difficulty: 2

77. What use is genetic mapping? Give one example where it has been of value to society.
    Ans: The cystic fibrosis gene was cloned by a positional method that first required detailed mapping. Cloning has led to some therapeutic interventions and holds hope for a cure.
    Difficulty: 1

78. Why does the formula for the map distance between a gene and the centromere count only half of the second division segregations?
    Ans: Because in each second division segregation ascus only four of the eight spores result from a crossover between the gene and the centromere.
    Difficulty: 3

79. Why does the formula for map distance between the two outside genes in a 3-point testcross count the double crossovers twice whereas the formula for the distance between an outside gene and the middle gene counts the double crossovers only once?
Ans: Because a double crossover is two single crossovers, one in one interval and one in the other. Since the distance between the outside genes includes both intervals, each crossover must be counted.
Difficulty: 2

80. When mapping three genes, why do geneticists prefer to do one three-point cross rather than three two-point crosses? Give two reasons.
Ans: 1. Three point crosses can establish gene order unambiguously. 2. Three point crosses account for double crossovers as well as single crossovers and so estimate map distances more accurately. 3. Three point crosses allow calculation of interference parameters.
Difficulty: 2

<u>Experimental Design and Interpretation of Data</u>

In *Drosophila*, the genes y, f and v are all X-linked. y f v females are crossed to wild-type males and the F1 females are test-crossed. The F2 are distributed as follows:

y f v	3210
y f +	72
y + v	1024
y + +	678
+ f v	690
+ f +	1044
+ + v	60
+ + +	<u>3222</u>
	10,000

81. Determine the linkage arrangement of these three genes and calculate the map distances.
Ans: y v f; y-v = 15 m.u., v-f = 22 m.u.; y-f = 37 m.u.
Difficulty: 3

82. Calculate the coefficient of coincidence.
Ans: c.c. = 0.4.
Difficulty: 3

83. State what a coefficient of coincidence of 0.5 would mean.
Ans: Half as many double crossovers were observed as would have been expected if crossovers in the two intervals were independent.
Difficulty: 3

84. Map distances between each pair of three linked genes L, M and N are measured by two point test crosses and determined to be: L-M = 20 cM, M-N = 25 cM, and L-N = 38 cM. Explain why the measured distance between the two farthest apart genes is not equal to the sum of the two shorter distances.
    Ans: The two point cross between L and N does not count the events in which crossovers occurred both between L and M and between M and N, but these are counted separately in the two smaller distances.
    Difficulty: 2

Dihybrid test crosses are made between each pairwise combination of the three genes C, D and E with the following results:

CD	222	CE	48	DE	198
Cd	280	Ce	455	De	289
cD	280	cE	445	dE	311
cd	218	ce	52	de	202
	1000		1000		1000

85. Which are the recombinant classes in these three crosses?
    Ans: CD and cd, CE and ce, DE and de
    Difficulty: 2

86. Give the genotypes of the dihybrids in these three crosses, showing linkage arrangements.
    Ans: Cd/cD, Ce/cE, and De/dD.
    Difficulty: 3

87. Draw the map for these three genes, indicating the order and the lengths of the intervals in map units.
    Ans: CED; C-E = 10 m.u., D-E = 40 m.u., C-D = 44 m.u.
    Difficulty: 3

88. What can you say about crossover interference based on this data set?
    Ans: Nothing, interference can be measured only in a 3-point cross.
    Difficulty: 2

89. Why do the map distances not add up?
    Ans: Because the C-D distance does not include the double crossovers.
    Difficulty: 2

Females heterozygous for the recessive second chromosome mutations pn, px and sp are mated to a male homozygous for all three mutations. The offspring are as follows:

px sp cn	1,410
px sp +	3,498
px + cn	1
px + +	11
+ sp cn	8
+ sp +	0
+ + cn	3,483
+ + +	1,489
	10,000

90. What is the genotype of the females that gave rise to these progeny?
    Ans: px sp cn+/px+ sp+ cn
    Difficulty: 3

91. Using the two point cross method, what are the distances among these three genes?
    Ans: px-sp = .2 m.u., px-cn = 30 m.u., cn-sp = 30.18 m.u.
    Difficulty: 3

92. Using the three point cross method, determine which gene is in the middle (which has to be flipped).
    Ans: px.
    Difficulty: 3

93. Which method provides more confidence as to the correct order of the genes?
    Ans: The 3-point method gives an unambiguous order because single crossovers can be distinguished from double crossovers. The 2-point method is less reliable because the difference between 30.0 and 30.18 is too small to provide a certain distinction.
    Difficulty: 2

94. Calculate the coefficient of coincidence and comment on its meaning.
    Ans: c.c.= 1/6 = .16; 6 double crossovers were expected but only one was observed, so interference is quite strong.
    Difficulty: 4

95. 50 map units is the maximum measurable map distance between two genes in a two-point cross, yet some human chromosomes are over 200 map units in length. Explain this discrepancy.
    Ans: Two point crosses do not count multiple crossovers, which is why 50 map units is the most that can be measured. But a 200 map unit chromosome has an average of four crossovers per meiosis.
    Difficulty: 2

In peas, tall (T) is dominant to short (t), red flowers (R) is dominant to white flowers (r), and wide leaves (W) is dominant to narrow (w). A tall, red, wide plant is crossed to a short, white, narrow plant and the progeny are as follows:

tall red wide	381
tall white wide	122
short red wide	118
short white wide	379
	1000

96. What is the genotype of the tall, red, wide parent?
    Ans: TtRrWW.
    Difficulty: 4

97. Give the linkage arrangements.
    Ans: T/t and R/r are linked and 24 map units apart. Linkage of W/w unknown.
    Difficulty: 3

98. A strain of Neurospora with the genotype MN is crossed with a strain that is mn. Half the progeny are MN and half are mn. Explain these results.
    Ans: Genes M and N are completely linked, so that no recombinants are produced.
    Difficulty: 3

A dihybrid test cross is made between the genes C and D with the following results:

CcDc × ccdd	
CD	222
Cd	280
cD	280
cd	218
	1000

99. You wish to test the hypothesis that the C and D genes are unlinked. What are the expected values for each class on the assumption that the genes are unlinked?
    Ans: 250 for each.
    Difficulty: 2

100. Calculate X^2 to test the hypothesis that the C and D genes are unlinked.
    Ans: $X^2 = 14.4$.
    Difficulty: 3

101. Determine the degrees of freedom and calculate the p value.
Ans: df = 3; $p < .01$.
Difficulty: 2

102. State what this p value means.
Ans: There is less than 1 chance in 100 that deviations from expected this large or larger could have been generated by chance, or, more simply, there is less than 1 chance in 100 that the C and D genes are unlinked.
Difficulty: 3

103. What would the outcome of this test have been if only 100 progeny had been counted and the same proportions observed?
Ans: X^2 would have been between 1.0 and 2.0, so p would have been much greater than .05; we would not have been able to rule out the hypothesis of no linkage.
Difficulty: 4

104. A female mouse from a true-breeding wild-type strain was crossed to a male mouse with apricot eyes (ap) and grey body (gy). The F1 mice were wild-type for both traits. When the F1 were interbred, the F2 were distributed as follows (200 total):
Females: all wild type

Males:	wild type	91
	apricot	11
	grey	9
	apricot, grey	89

Explain these results.
Ans: ap and gy are X-linked and 10 map units apart.
Difficulty: 2

Suppose the map for a particular human chromosome interval is: a-1-b-1-c-1-d-1-e-1-f, where the numbers indicate map distances.

105. In a man heterozygous for all six genes, what fraction of his sperm would be recombinant in the a-f interval?
Ans: 5%.
Difficulty: 3

106. In a man of the genotype ABCDEF/abcdef, what event would give rise to a sperm with the genotype abcdeF?
Ans: A crossover between the E/e and F/f loci.
Difficulty: 2

In Neurospora the cross AB × ab was made and 100 asci were scored:

I	II	III	IV	V	VI	VII
A B	A B	A B	A B	A B	A b	A b
A B	A B	A B	A B	A B	A b	A b
A B	A b	a B	a b	a b	a B	A b
A B	A b	a B	a b	a b	a B	A b
a b	a B	A b	A B	A b	A b	a B
a b	a B	A b	A B	A b	A b	a B
a b	a b	a b	a b	a B	a B	a B
a b	a b	a b	a b	a B	a B	a B
39	16	32	2	6	4	1

107. Which classes are the PDs, NPDs & Ts?
    Ans: PD – I & IV; NPD – VI, VII; T - II, III & V.
    Difficulty: 2

108. Which class(es) are second division segregations?
    Ans: For A – III, IV, V and VI; for B – II, IV, V, and VI.
    Difficulty: 2

109. What is the map of these genes and any linked centromere(s)?
    Ans: A-cent-B; A-cent = 22; cent-B = 14; A-B = 32.
    Difficulty: 3

110. How do we know that crossing-over is reciprocal?
    Ans: Because when a recombinant spore class is present in an ascus, the reciprocal recombinant class is generally present as well.
    Difficulty: 3